Bihar Engineering University, Patna

B.Tech. 8th Semester Examination, 2024

Course: B.Tech. *Time: 03 Hours*

Code: 100802 *Subject: Control Systems Design* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer of the following (any seven questions only):*** **[2 x 7 = 14]**

(a) Which system exhibits the initiation of corrective action only the output gets affected?

(i) Feedback (ii) Feed forward

(iii) Both A and B (iv) None of these

(b) What is the value of steady state error in closed loop control systems?

(i) Unity (ii) Infinity

(iii) Zero (iv) Unpredictable

(c) PID controller

(i) increases steady state error

(ii) transient response becomes poorer

(iii) decreases steady state error and improves stability

(iv) rise time decreases

(d) Which of the following is an affect of lag compensation?

(i) Increases the effect of noise (ii) decrease bandwidth

(iii) increases stability margin (iv) improves transient response

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(e) The Laplace transform $F(s)$ of $f(t) = tu(t)$ is

(i) $\frac{1}{s}$ (ii) $\frac{1}{s^2}$

(iii) $\frac{1}{s^3}$ (iv) 1

(f) For a state space model given by equations $\dot{x}_1 = 2x_2$; $\dot{x}_2 = -3x_1 - 5x_2 + e^{-t}$ with initial conditions $x_1(0) = 2$; $x_2(0) = 1$. Eigenvalues are given by

(i) -1, -2 (ii) -2, -3

(iii) -3, -4 (iv) None of the above

(g) The system given as is $T(s) = \frac{K(s+3)}{s(s+1)(s+2)}$

(i) Type 0 System (ii) Type 3 System

(iii) Type 1 System (iv) Type 2 System

(h) The phase margin is the change in open-loop phase shift required at ..... to make the closed-loop system unstable

(i) unity input signal (ii) unity feedback

(iii) unity gain (iv) transfer function

(i) In Nyquits diagram we explore mapping of .....

(i) Contours (ii) Configuration Space

(iii) Limit cycles (iv) Phase portraits

(j) Write mason's gain formula.

Q.2 Discuss in detail:

(a) PID Controllers **[7]**

(b) Lag-Lead Compensators **[7]**

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Q.3 Discuss following time domain specifications: **[14]**

a) Maximum Overshoot
b) Settling Time
c) Rise Time
d) Peak Time.

Find out the above mentioned specifications for the given transfer function.

$G(s) = \frac{50}{s^2+15s+70}$

Q.4 What do you mean by controllability and Observability explain with supporting mathematical formulations. **[14]**

Q.5 What do you mean state variables? Write equations for state space representation. Discuss in detail the solution of state equation. Also give formulation for diagonalization of state transition matrix. **[14]**

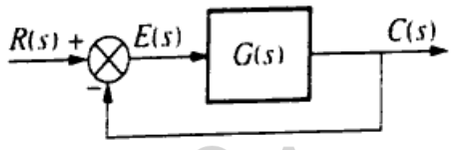
Q.6 (a) Explain in detail of effect of addition of pole and zero on system performance. **[7]**

(b) Determine the number of right-half-plane poles in the closed-loop transfer function **[7]**

$T(s) = \frac{10}{s^4+7s^3+5s^2+40s+5}$

Q.7 Sketch the root locus and its asymptotes for a unity feedback system that has the forward transfer function $G(s) = \frac{K}{(s+3)(s+5)(s+6)}$ **[14]**

Q.8 Draw the bode plot of the system shown in figure 1 where $G(s) = \frac{K(s+2)}{s(s+1)(s+3)}$ **[14]**



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Q.9 Write short notes on ***any two*** of the following: **[14]**

(i) Effect of various non-linearities on system performance.
(ii) Difference between transient and frequency response.
(iii) Feedback and feedforward compensator design.

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